Lab 1: FPGA-based Mental Binary Math Game
User Manual

Tru Nguyen
PSID: 1911032
ECE 6370

Description

A game that can be played by two players on the DE0-CV FPGA board. Each player has one set of four slide switches on the board. Player 1 switches are the four rightmost switches. Player 2 switches are the four leftmost switches (SW9-SW6). Player 1 and Player 2' numbers are displayed on the rightmost display and the second to leftmost display respectively. The sum of the two players' numbers is displayed on the third display from the right. The leftmost or rightmost LED is lit if a matching or non-matching condition occurs respectively.

The rule of the game is that Player 1 sets their number using the switches then Player 2 needs to set a number (one-time only) that the sum of both numbers equals 1111 (binary) or F on display without looking at Player 1's switches. If Players 2 gets the right number, they win the round. If Player 2 gets the wrong number, Player 1 wins the round. The players swich roles after every round as Player 2 now sets a number for Player 1 to match.

Binary (switches)	Hexadecimal (display)	Decimal
0000	0	0
0001	1	1
0010	2	2
0011	3	3
0100	4	4
0101	5	5
0110	6	6
0111	7	7
1000	8	8
1001	9	9
1010	A	10
1011	B	11
1100	C	12
1101	D	13
1110	E	14
1111	F	15

Table 1: Hexadecimal (on display) and binary (switches) conversion to decimal

FPGA Board Overview

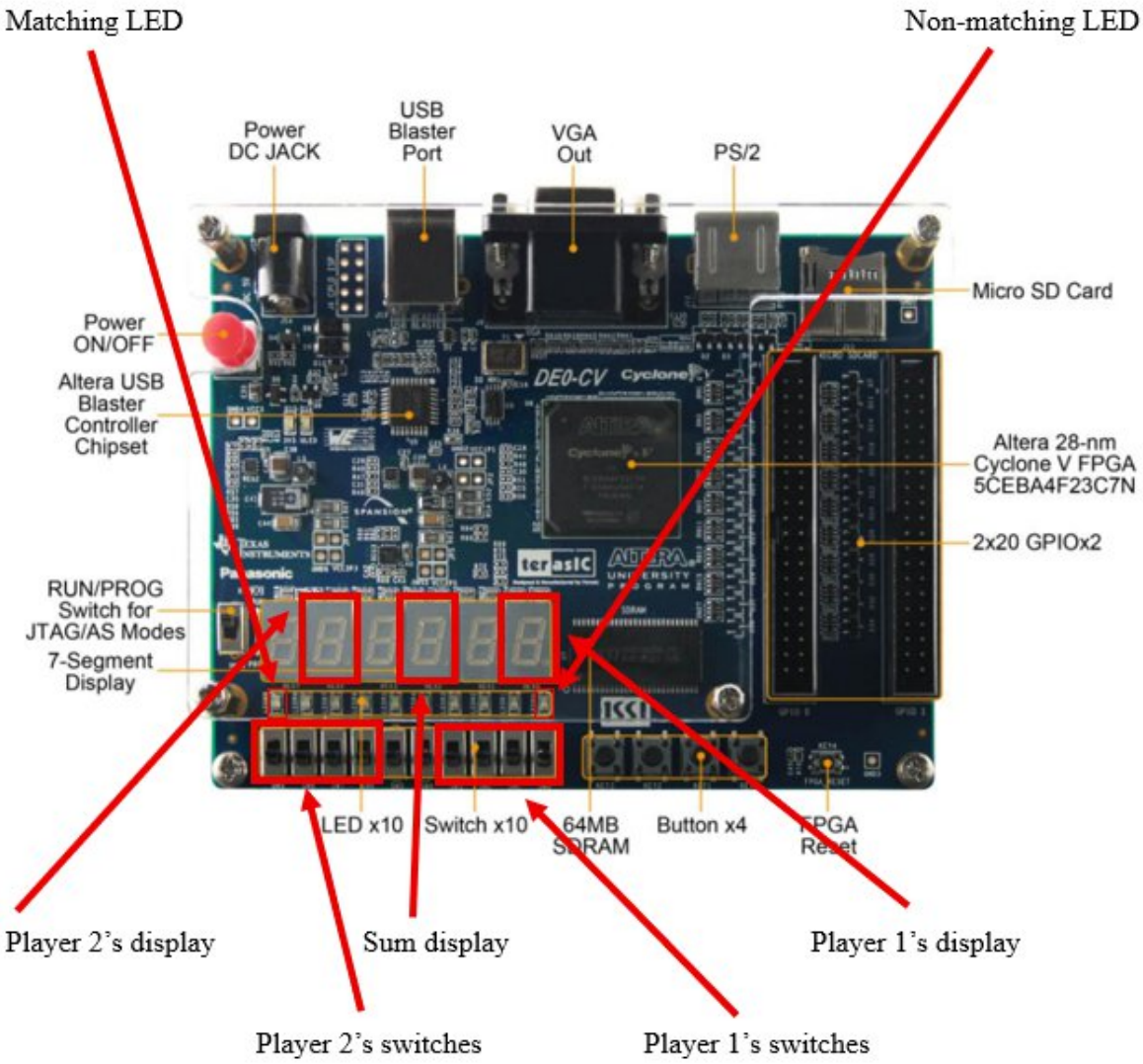

Figure 1: DE0-CV FPGA board with annotated Players' switches, displays, and LEDs

Step-by-step Game Rule

1. Win round

Step 1: FPGA is powered and in default settings showing both players' numbers are 0000 (switches) or 0 (hex) displayed on respective 7-segment displays. The rightmost LED is lit depicting non-matching condition while the leftmost LED is turned off.

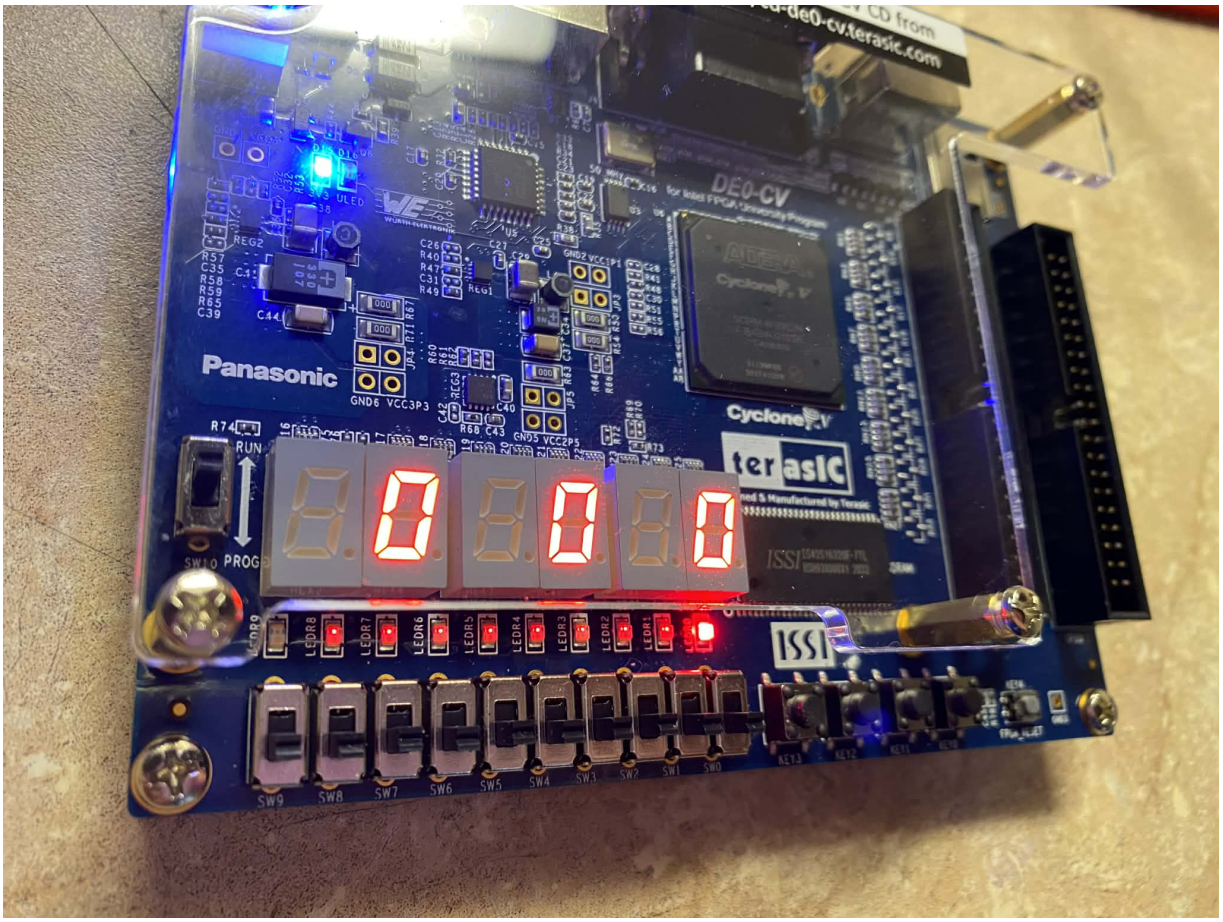

Figure 2: FPGA default settings

Step 2: Player 1 enters their number which is 0110 (6 in hex) and is shown on the right display. The middle display depicts the sum of two numbers which is 0110 (6 in hex) in this case because Player 2 hasn't entered their number. The rightmost LED is still lit because of non-matching condition.

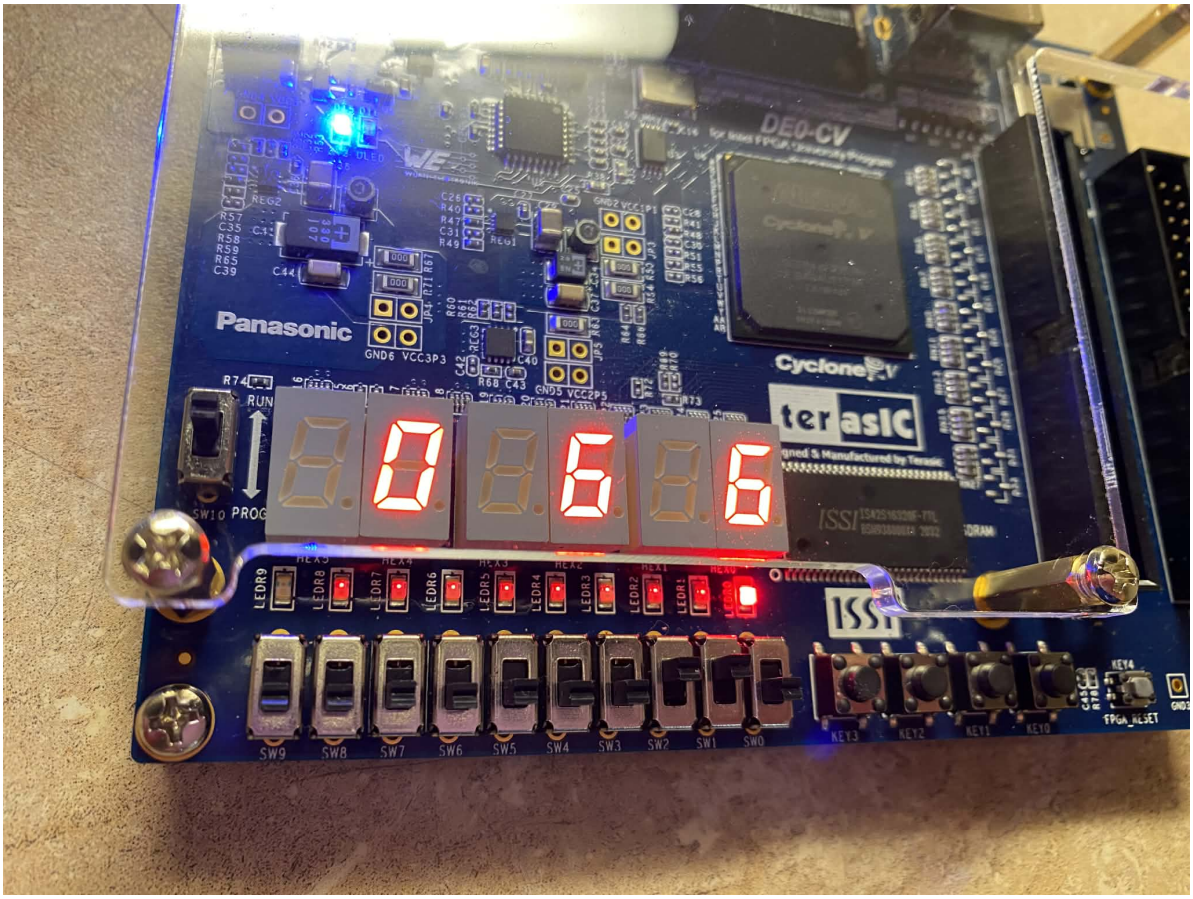

Figure 3: Player 1 number entered

Step 3: Player 2 enters their number which is 1001 (9 in hex) illustrated on the left display. The sum is now 1111 (F in hex). The leftmost LED is now lit up indicating matching condition while the rightmost LED is off. Player 2 is considered the winner of this round.

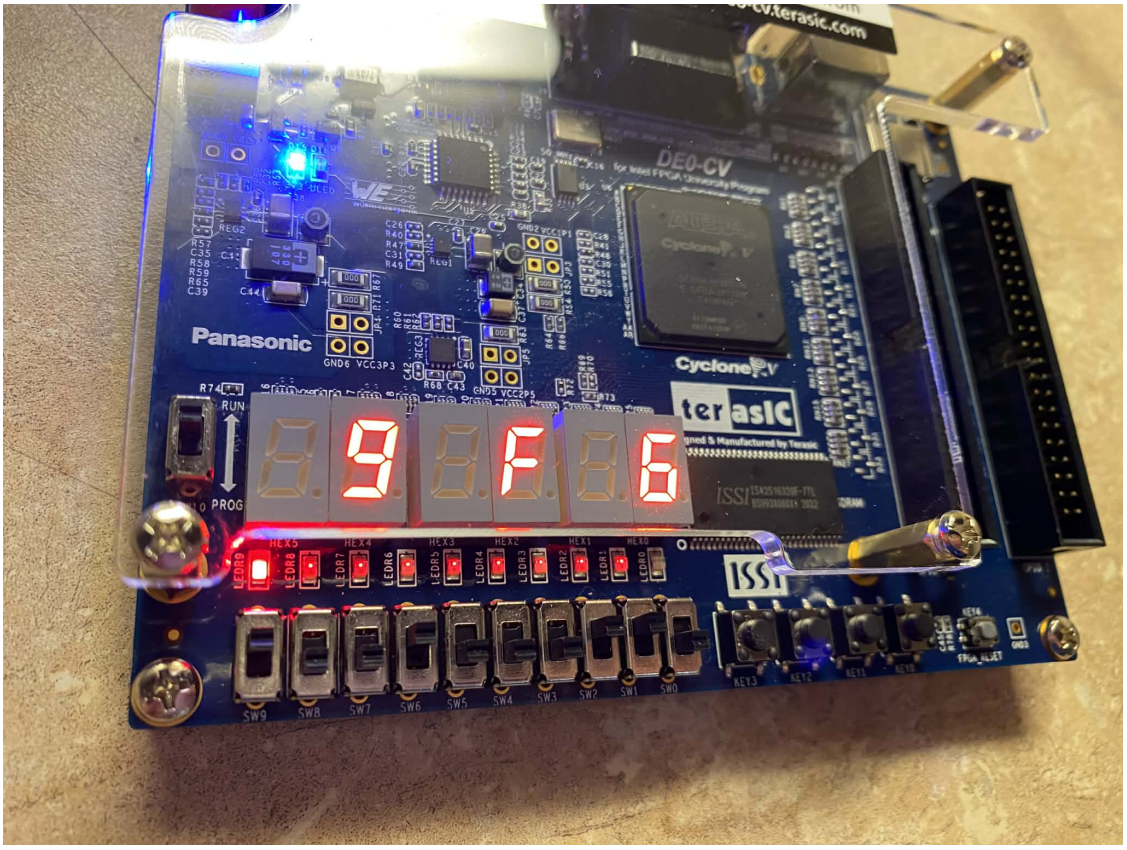

Figure 4: Player 2 number entered and matching condition shown on the FPGA

2. Lose round

Step 1: FPGA is powered and in default settings showing both players' numbers are 0000 (switches) or 0 (hex) displayed on respective 7-segment displays. The rightmost LED is lit depicting non-matching condition while the leftmost LED is turned off.

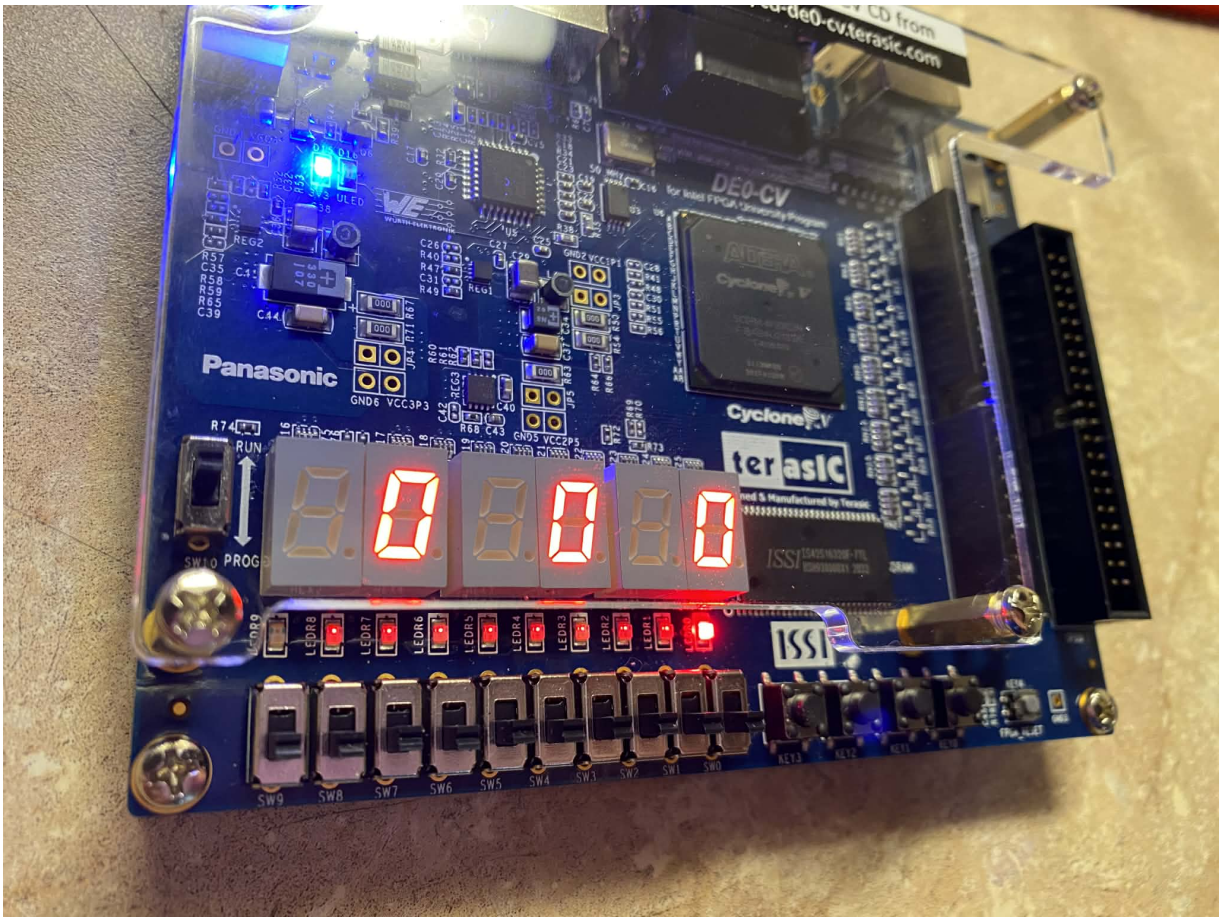

Figure 5: FPGA default settings

Step 2: Player 1 enters their number which is 0110 (6 in hex) and is shown on the right display. The middle display depicts the sum of two numbers which is 0110 (6 in hex) in this case because Player 2 hasn't entered their number. The rightmost LED is still lit because of non-matching condition.

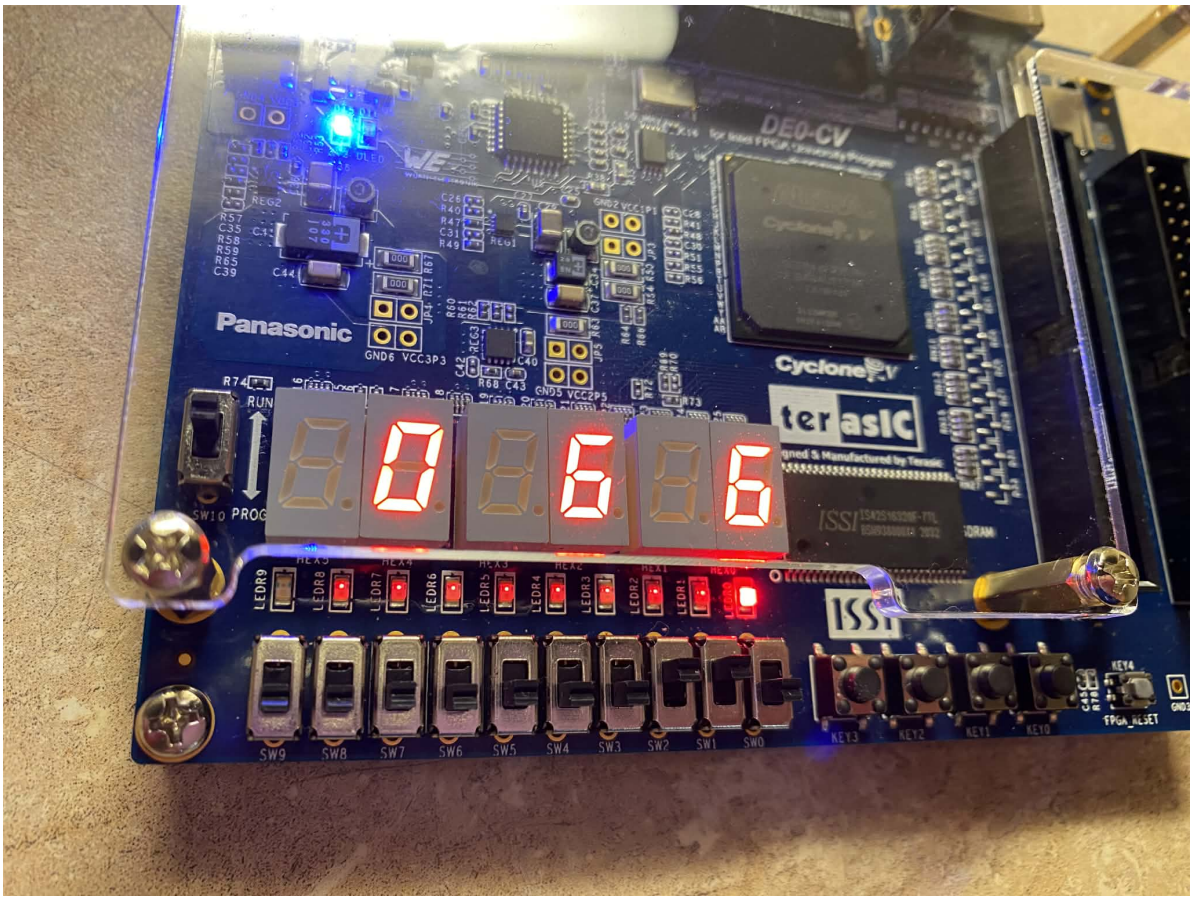

Figure 6: Player 1 number entered

Step 3: Player 2 enters their number which is 0011 (3 in hex) illustrated on the left display. The sum is now 1001 (9 in hex). The rightmost LED is now lit up indicating non-matching condition while the leftmost LED is off. Player 2 loses the round and Player 1 is considered the winner of this round.

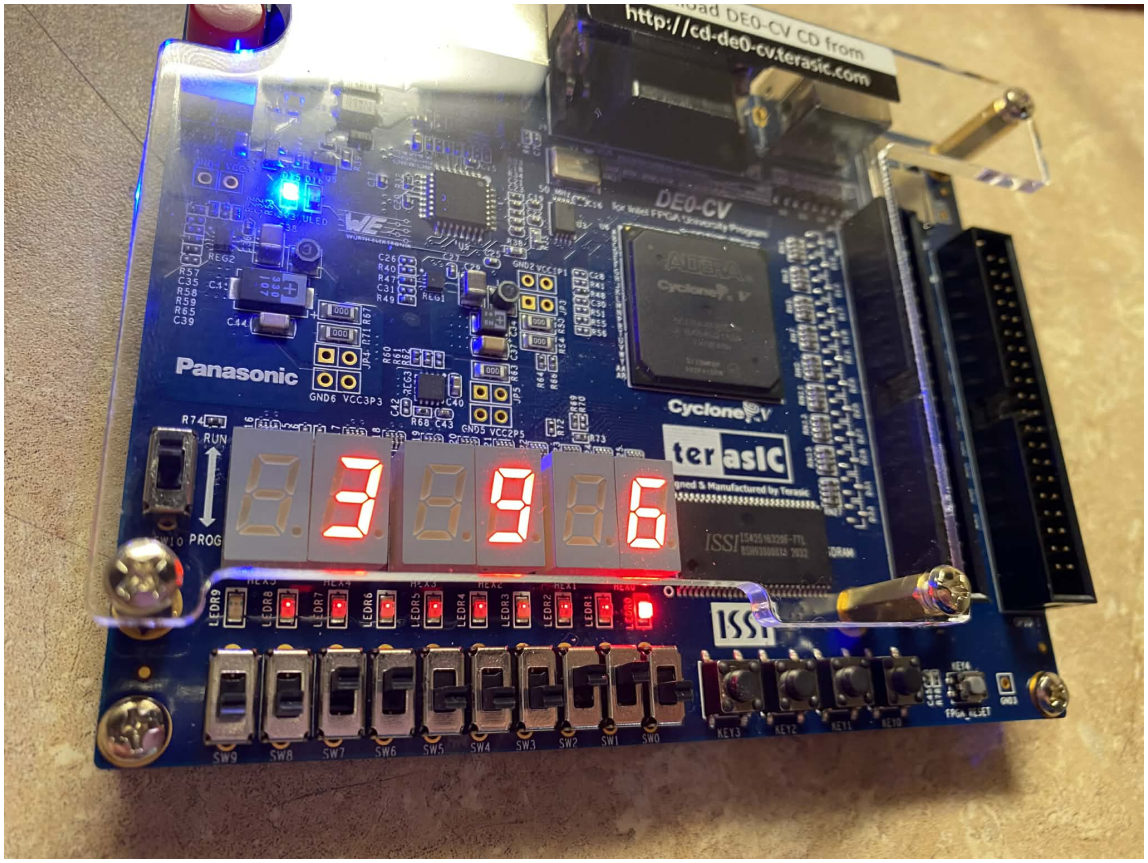

Figure 7: Player 2 number entered and non-matching condition shown on the FPGA